

Assignment 2: Nmap and VirusTotal Analysis

1. Nmap Commands Used:

Command 1: Basic TCP Scan:

`nmap scanme.nmap.org`

This command performs a basic TCP scan against the target domain scanme.nmap.org. It is used to identify open ports and determine whether the host is reachable. The scan revealed several open ports and confirmed that the host is active. This initial scan provides an overview of exposed network services running on the target system.

```
PS C:\Ship\Info> nmap --version
Nmap version 7.98 ( https://nmap.org )
Platform: i686-pc-windows-windows
Compiled with: nmap-liblua-5.4.8 openssl-3.0.17 nmap-libssh2-1.11.1 nmap-libz-1.3.1 nmap-libpcap-1.0.4 Npcap-1.83 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: iocp poll select
PS C:\Ship\Info> nmap scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-02 10:41 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.23s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 990 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
23/tcp    filtered telnet
80/tcp    open  http
445/tcp    filtered microsoft-ds
2323/tcp   filtered 3d-nfsd
3005/tcp   filtered deslogin
6881/tcp   filtered bittorrent-tracker
9929/tcp   open  nping-echo
31337/tcp  open  Elite
50002/tcp  filtered iiimf

Nmap done: 1 IP address (1 host up) scanned in 12.45 seconds
PS C:\Ship\Info> |
```

Command 2: Service Version Detection (-sV):

`nmap -sV scanme.nmap.org`

The -sV option enables service version detection. This scan was used to identify the specific software versions running on the open ports discovered in the basic scan. It revealed services such as OpenSSH and Apache HTTP Server along with their respective version information. This helps in understanding the exact software exposed to the network and is important for vulnerability assessment.

```

Nmap done: 1 IP address (1 host up) scanned in 12.45 seconds
PS C:\Ship\Info> nmap -sV scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-02 10:52 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.30s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 990 closed tcp ports (reset)
PORT      STATE    SERVICE      VERSION
22/tcp    open     ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
23/tcp    filtered telnet
80/tcp    open     http         Apache httpd 2.4.7 ((Ubuntu))
445/tcp    filtered microsoft-ds
2323/tcp  filtered 3d-nfsd
3005/tcp  filtered deslogin
6881/tcp  filtered bittorrent-tracker
9929/tcp  open     nping-echo   Nping echo
31337/tcp open     tcpwrapped
50002/tcp filtered iiimsf
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 32.42 seconds
PS C:\Ship\Info>

```

Command 3: Default Script Scan (-sC):

`nmap -sC scanme.nmap.org`

The `-sC` option runs Nmap's default script engine to perform additional enumeration on open ports. This scan provides more detailed information about services, such as banner details and basic security configurations. It helps in gathering deeper insights beyond simple port and version detection and may reveal potential misconfigurations.

```

PS C:\Ship\Info> nmap -sC scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-02 11:03 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.22s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 990 closed tcp ports (reset)
PORT      STATE    SERVICE
22/tcp    open     ssh
| ssh-hostkey:
|   1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
|   2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
|   256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
|_  256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
23/tcp    filtered telnet
80/tcp    open     http
|_ http-title: Go ahead and ScanMe!
|_ http-favicon: Nmap Project
445/tcp    filtered microsoft-ds
2323/tcp  filtered 3d-nfsd
3005/tcp  filtered deslogin
6881/tcp  filtered bittorrent-tracker
9929/tcp  open     nping-echo
31337/tcp open     Elite
50002/tcp filtered iiimsf

Nmap done: 1 IP address (1 host up) scanned in 35.12 seconds
PS C:\Ship\Info>

```

2. Scan Results:

A) Open Ports Table:

Port	State	Service
22	Open	SSH
80	Open	HTTP
9929	Open	Nping-echo
31337	Open	Elite

B) Services and Versions:

Port 22 (SSH) was running OpenSSH 6.6.1p1 on Ubuntu Linux.

Port 80 (HTTP) was running Apache httpd 2.4.7 (Ubuntu).

Port 9929 was identified as the Nping echo service.

Port 31337 was identified as the Elite service.

Service version detection is important because it reveals the exact software and version running on the target system. This information helps in identifying known vulnerabilities associated with outdated or misconfigured services.

3. VirusTotal Findings:

A) Reputation and Detection Results

VirusTotal analysis of scanme.nmap.org shows that the domain is marked as clean by multiple security vendors. The reputation score is low risk, and no major antivirus engines flagged the domain as malicious.

The last analysis statistics indicate:

Malicious: 0

Suspicious: 0

Harmless: Majority of vendors

Undetected: Remaining vendors

This suggests that the domain is safe and not associated with malware or malicious activity.

0

/ 93

Community Score

1

At least 8 detected files communicating with this domain

Reanalyze More

scanme.nmap.org

nmap.org

advice information technology downloads top-1M

Registrar

Dynadot Inc

Creation Date

27 years ago

Last Analysis Date

14 hours ago

DETECTION

DETAILS

RELATIONS

COMMUNITY 10

Security vendors' analysis

Do you want to automate checks?

Abusix	✓ Clean	Acronis	✓ Clean
ADMINUSLabs	✓ Clean	AILabs (MONITORAPP)	✓ Clean
AlienVault	✓ Clean	alphaMountain.ai	✓ Clean
Antiy-AVL	✓ Clean	benkow.cc	✓ Clean
BitDefender	✓ Clean	Blueliv	✓ Clean

B) Domain and IP Intelligence

VirusTotal provides passive intelligence about the domain, including:

Registrar: Dynadot Inc

Creation Date: Approximately 27 years ago

Associated IP: 45.33.32.156

Categories: Information Technology / Business / Computer Security

Popularity rank data available

The domain has historical DNS records and passive DNS resolutions showing consistent and legitimate infrastructure usage.

0

/ 93

Community Score

1

At least 8 detected files communicating with this domain

Reanalyze More

scanme.nmap.org

nmap.org

advice information technology downloads top-1M

Registrar

Dynadot Inc

Creation Date

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DETECTION

DETAILS

RELATIONS

COMMUNITY 10

Categories

BitDefender	advice
Forcepoint ThreatSeeker	information technology
Sophos	information technology
Xcitium Verdict Cloud	downloads
alphaMountain.ai	Business/Economy, Information Technology, Information/Computer Security (alphaMountain.ai)

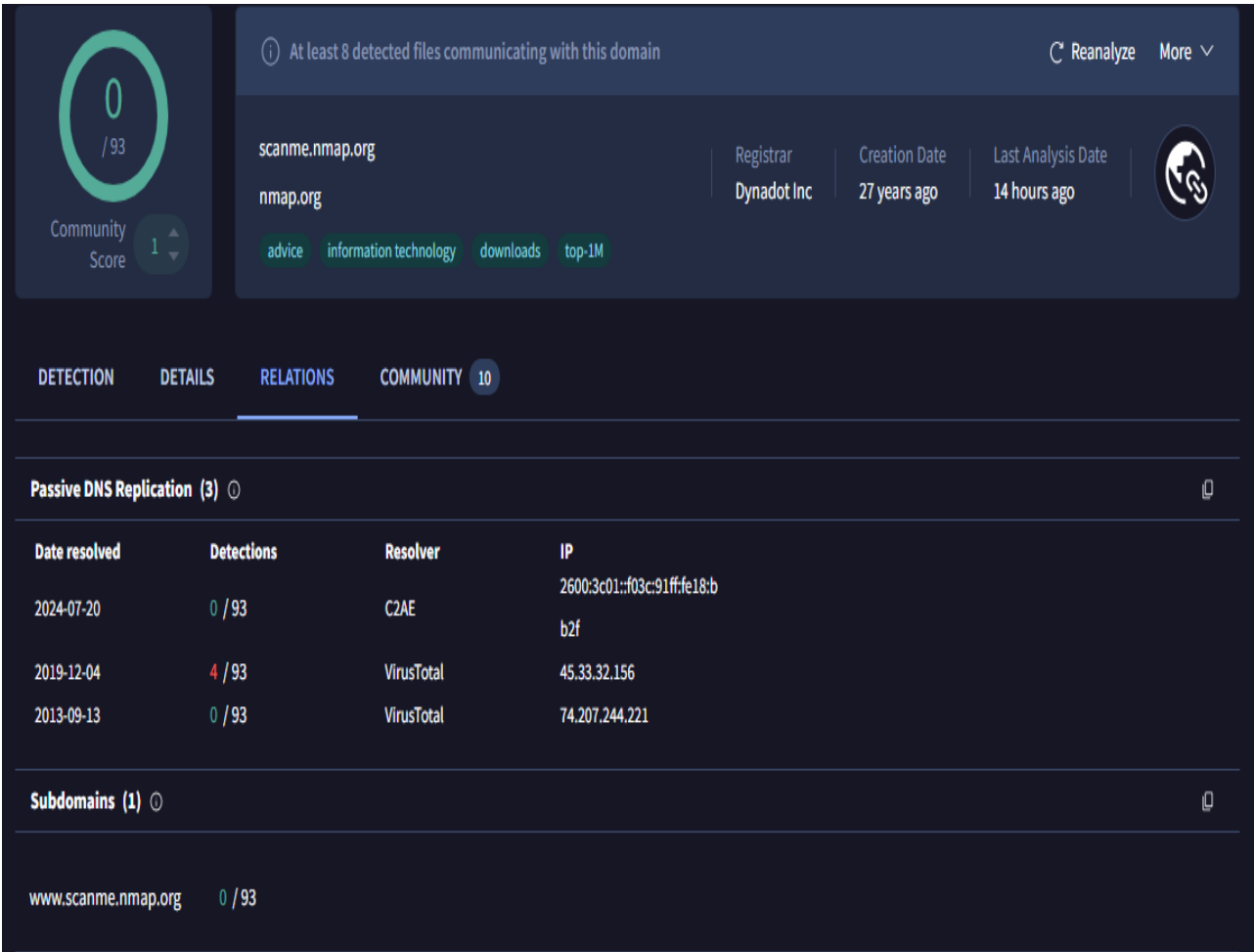
Popularity ranks

Rank	Position	Ingestion Time
Cisco Umbrella	965810	2026-01-01 14:38:22 UTC

Last DNS records

C) Historical / Contextual Data

Passive DNS replication records show previous IP associations over time. No abnormal or malicious redirections were observed. The domain maintains stable infrastructure and appears to be used for legitimate testing purposes by the Nmap project.



D) API-Based Analysis (Python)

The VirusTotal API was used to programmatically retrieve domain intelligence. The script extracted:

Reputation score

Last analysis statistics

This demonstrates how passive intelligence can be automated using the VirusTotal API.

```
1 import requests
2 import os
3 from dotenv import load_dotenv
4
5 load_dotenv()
6 API_KEY = os.getenv("API_KEY")
7
8 BASE_URL = "https://www.virustotal.com/api/v3"
9
10 headers = {
11     "x-apikey": API_KEY
12 }
13
14 def get_data(endpoint):
15     url = f"{BASE_URL}/{endpoint}"
16     response = requests.get(url, headers=headers)
17
18     if response.status_code == 200:
19         return response.json()
20     else:
21         print("Error:", response.status_code)
22         return None
23
24
25 def analyze_domain(domain):
26     endpoint = f"domains/{domain}"
27     data = get_data(endpoint)
28
29     if data:
30         attributes = data["data"]["attributes"]
31
32         print("Reputation:", attributes.get("reputation"))
33         print("Last Analysis Stats:", attributes.get("last_analysis_stats"))
34
35
36 analyze_domain("scanme.nmap.org")
```

PROBLEMS	OUTPUT	DEBUG CONSOLE	TERMINAL	PORTS
			PS C:\Ship\Info> & C:\Users\praga\AppData\Local\Python\pythoncore-3.14-64\python.exe c:/Ship/Info/main.py API Key not found! PS C:\Ship\Info> & C:\Users\praga\AppData\Local\Python\pythoncore-3.14-64\python.exe c:/Ship/Info/main.py Reputation: 1 Last Analysis Stats: {'malicious': 0, 'suspicious': 0, 'undetected': 33, 'harmless': 60, 'timeout': 0} PS C:\Ship\Info> & C:\Users\praga\AppData\Local\Python\pythoncore-3.14-64\python.exe c:/Ship/Info/main.py Reputation: 1 Last Analysis Stats: {'malicious': 0, 'suspicious': 0, 'undetected': 33, 'harmless': 60, 'timeout': 0} PS C:\Ship\Info>	

4. Port & Service Analysis

Port 22 (SSH) is used for Secure Shell remote access. It allows administrators to securely manage systems over an encrypted connection. If improperly configured, SSH may be vulnerable to brute-force attacks or weak authentication mechanisms.

Port 80 (HTTP) is used for web traffic. It serves web pages over the internet. If the web server software is outdated or misconfigured, it may be vulnerable to web-based attacks such as cross-site scripting (XSS) or directory traversal.

Port 9929 is associated with the Nping echo service. This service is typically used for testing and debugging network connectivity. While not inherently dangerous, unnecessary exposure of test services may increase attack surface

Port 31337 is identified as the Elite service. This port number is historically associated with backdoor services in some systems. Although in this case it belongs to the Nmap test server, unusual high-numbered ports should always be analyzed carefully in real-world environments.

5. Vulnerability Reasoning:

Based on the service and version information obtained from the Nmap scan, potential vulnerabilities can be inferred. OpenSSH 6.6.1p1 and Apache httpd 2.4.7 are older software versions, and older versions may contain publicly known vulnerabilities if not properly patched.

For example, outdated SSH services may be vulnerable to brute-force attacks or cryptographic weaknesses. Similarly, older Apache versions may be susceptible to web-based attacks such as denial-of-service vulnerabilities or misconfiguration issues.

Although no exploitation was performed, identifying service versions helps security analysts assess potential risks and determine whether updates or configuration changes are required.

6. Conclusion:

This assignment provided practical exposure to both active scanning and passive intelligence gathering techniques. Using Nmap, open ports and running services were identified on the target system, along with their respective versions. This demonstrated how network scanning helps in identifying exposed services and potential security risks.

Using VirusTotal, passive intelligence about the domain was gathered without directly interacting with the target system. The findings confirmed that the domain is legitimate and not associated with malicious activity. The API-based analysis further demonstrated how security intelligence can be automated using Python.

Overall, this exercise highlighted the importance of combining active scanning tools like Nmap with passive intelligence platforms like VirusTotal to perform comprehensive security analysis.